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CONDUCTED BY R. CHAMBERS (SECUNDUS)

No. 13.—VOL. I.

SATURDAY, MARCH 29, 1884.

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INTERNATIONAL FORESTRY EXHIBITION, EDINBURGH, 1884.

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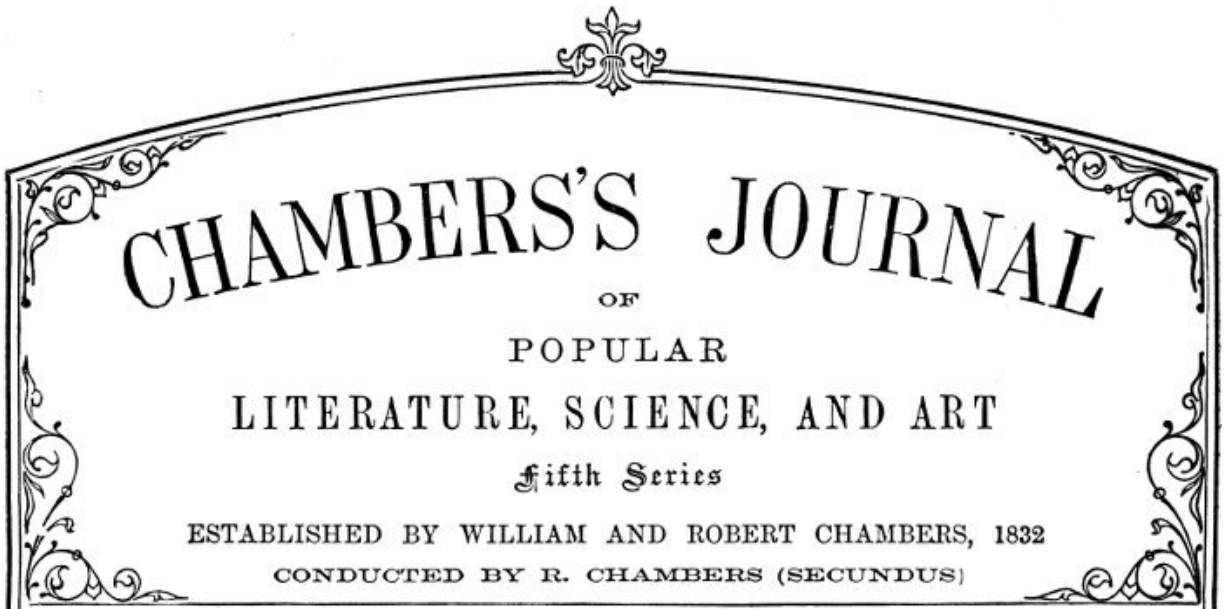
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It is not now our object to show how the government of India has suffered in the interval by the want of a proper system of forest training in Great Britain. Waste and spoliation went on, of course, uncontrolled. But we think that the *raison d'être* of a Forestry Exhibition will now be tolerably apparent to at least the majority of our readers. Indeed, the wonder is that Great Britain has so long remained quiescent under the implied reproach of neglecting what is not only a useful but a profitable branch of estate management. This reproach, which had long weighed on the minds of all those who had the good of the country at heart, at length found public expression at the meetings of the principal Societies of Scotland who represent the landed interest of the country, and resolutions were passed pledging their members to the support of a Forestry Exhibition.

Meanwhile, the great success of the Fisheries Exhibition in London had induced the executive Committee there to try and achieve for other

industries similar benefits to what they had conferred on the fishermen of England. And they, too, pitched upon forestry as a branch of science well worthy of encouragement. But when it was represented to them that the same idea, first mooted in Scotland, had already assumed practical shape there, they courteously gave way, and conceded to Scotland the well-deserved right of holding in her capital the first Forestry Exhibition of Great Britain.

Nearly all the foreign powers and the representatives of our colonial and Indian empire are to be found in the list of those who have joined the undertaking as members. And the following letter, which has been sent to our diplomatic representatives abroad, rightly expresses the consensus of official and public opinion on the merit of the undertaking:

(CIRCULAR-COMMERCIAL.)

FOREIGN OFFICE, *October 27, 1883.*

The attention of Her Majesty's government has been directed to a project for an International Exhibition of Forestry to be held in Edinburgh in the summer of 1884, the organisers of which are desirous of securing the co-operation therein of such foreign countries as the matter may concern. There is reason to believe that the proposed Exhibition, for which the necessary funds have been guaranteed, will be influentially and ably supported. The object is one which in the opinion of Her Majesty's government deserves every encouragement, scientific forestry having hitherto been much neglected in this country; and I have therefore to request that you will bring the Exhibition in question to the notice of the government to which you are accredited, as being one in which their participation might be attended with advantage to both countries. I inclose for communication to the proper quarters copies of programme and other documents connected with the proposed Exhibition, which have been supplied by the Committee.

I am, with great truth, your most obedient humble servant,

Signed (for Earl Granville)

EDMOND FITZMAURICE.

With this letter, we may fitly close the contemplation of the causes which have led to the idea of a Forestry Exhibition being held in Great Britain. They are, in fact, briefly summed up in the short but comprehensive dictum, which, we fear, cannot be contradicted or gainsayed, 'scientific forestry having hitherto been much neglected in this country.' And the inverse of this proposition leads us by no indirect steps to the consideration of the good results which may be expected to accrue from the Exhibition, if it is successfully conducted.

To the capital of Scotland, a country lying between the two great fields of the 'lumbering' interest of the world—the one in Northern Europe, and the other on the continent of America, the results, if only from the influx of visitors, whether these are scientifically disposed or otherwise, can hardly fail to be beneficial. But there are wider interests involved. The landed proprietor anxious to utilise his present wastes and to make up for deficient rents by profitable planting—the political economist inquiring into new sources of revenue—the botanist uncertain of the right names and uses of some of his specimens of timber or of flowers—the geologist, the sportsman, and the naturalist, will find here a common ground of instruction and amusement. For we may hope to see gathered together the forest products of the world, carefully examined and authentically named; the various descriptions of machines used in different countries for preparing timber for constructive purposes; the timber slips placed on the hills, the sluices, dams, and embankments formed on the rivers for the transporting of wood by land or by water; the mechanical appliances used for moving growing trees, and the saw-mills for cutting them into sections when felled. Here, too, will be exhibited the various textile fabrics manufactured from bark; materials for the making of paper; tanning and dyeing substances; drugs and spices; gums, resins, wood-oils and varnishes. Another section will embrace botanical specimens, fungi and lichens, forest entomology and natural history; with fossil plants and the various trees found in bogs.

The literature of the subject will be illustrated by the Reports of Forest schools, the working plans of plantations, which show the age of the various woods on an estate, and the stage of growth at which they may most profitably be thinned or felled. Remarkable or historical trees will be represented by paintings, photographs, and drawings; and there will be sketches of the usual forest operations.

Collections of forest produce, specially illustrating the sources of supply, and the methods of manufacture in different provinces, with accompanying Reports, are solicited by the Committee. And essays on all subjects touching on the value of growing trees or timber are invited to competition for prizes. Here, again, is opened a very wide field of useful inquiry for all those interested in the planting of woods in our own or foreign countries; for the cultivators of cinchona and other barks in our Crown colonies; for wood-engravers, whose supply of hard wood for the purposes of their trade is now very limited; for ship-builders, anxious to get a substitute for teak, or to obtain an increased supply of that most useful timber; and for all who use wood or forest produce in any of the many forms of manufacture in which they are applied.

We may not enter into any further categorical enumeration of the purposes and objects of the Forestry Exhibition of 1884; for, if the contemplation of the great cause which primarily led to the idea of the undertaking has brought us insensibly to the enumeration of the good that may be expected to accrue from its successful issue, it seems needless to insist that the probable results of that success will benefit the commercial interests and the scientific knowledge of the world at large. [Particulars of the Exhibition may be obtained from Mr George Cadell, secretary, 3 George IV. Bridge, Edinburgh.]

BY MEAD AND STREAM.

CHAPTER XX.—PAVED WITH GOLD.

IT was not probable that Mr Hadleigh would object to his son being endowed with a fortune by a wealthy uncle, whatever might be his feelings towards the donor. He would no doubt have been ready with congratulations if the endowment had come from any other quarter. As the case stood, Philip anticipated some difficulty in reconciling him to the arrangement, unless he should succeed in making the two men forget and forgive that old feud. However, there would be time enough to consider these details after the consultation with the solicitors.

He found Mr Hawkins and Mr Jackson together in the senior partner's room—a rare circumstance for any client to find them so, for acting separately, they might cancel or amend opinions after private conference without loss of prestige. On the present occasion Philip's affairs had been the subject of discussion.

'Let me offer you my best congratulations, sir,' said Mr Hawkins, a thin, grave-looking old gentleman in speckless black broadcloth, and with gold-mounted glasses on his prominent nose.

'Accept the same from me, Mr Hadleigh,' interjected Jackson. He was a sharp gentleman of middle age, with small mutton-chop whiskers, and dressed in the latest City fashion—for there is a City fashion, designed apparently to combine the elegance of the west end with a suggestion of superhuman 'cuteness.'

'Thank you, both. I must be a lucky fellow when you say so.'

'In the course of my experience,' said Mr Hawkins solemnly, 'I have never known a young man start in life under such favourable auspices. We wish you success, and we believe you will find it difficult to fail.'

'It is wonderful what a fool can do,' said Philip, laughing; 'but I will try not to fail. At present, I am a little in the dark as to the terms of the proposed arrangement, and Mr Shield referred me to you for the particulars.'

‘The particulars are simple,’ the lawyer proceeded slowly, as he turned over a number of papers on which various notes were written. ‘In the first place, I have great—very great—pleasure in informing you that a sum of fifty thousand pounds has been paid into your credit at the Universal Bank; and a second sum of the same amount will be at your command whenever you may have occasion for it, provided Mr Shield is satisfied with the manner in which you have disposed of the first sum.’

‘This is scarcely the kind of arrangement I expected. I had a notion that it was to be a partnership,’ said Philip.

‘The arrangement is so simple and so complete, Mr Hadleigh, that you will have no difficulty in comprehending every detail presently.’ Mr Hawkins went on leisurely, as if he enjoyed prolonging the agreeable statement he had to make. Mr Jackson nodded his head at the close of every sentence, as if thereby indorsing it. ‘We have often read in story-books of rich uncles coming home to make all their friends comfortable. You have the exceptional experience of finding a rich uncle in reality—one who is resolved to pave your way with gold, as I may express it.’

‘But what does he want me to do with all this money?’ asked Philip, desirous of bringing the loquacious old gentleman to the point.

Mr Hawkins was not to be hurried. Like a connoisseur with a glass of rare wine, he was bent on making the most of it. Every symptom of eagerness on Philip’s part added zest to the palate; and he was graciously tolerant of his client’s impatience.

‘As regards the partnership, that will come afterwards. In the meantime, he desires you to consider this handsome fortune as absolutely at your own disposal. He imposes no conditions. You are free to give up all thought of profession or trade, and to live as you please on the income of this capital, or on the capital itself, if you are so inclined.’

‘That, of course, is nonsense. He must wish me to do something.’

‘Certainly; and although he imposes no conditions, he has expressed two wishes.’

‘And what are they?’

Mr Hawkins polished his eye-glasses and consulted his notes. Mr Jackson nodded his head pleasantly, as if he were saying: 'Now it is coming, you lucky dog.'

'The first is,' Mr Hawkins went on, 'that you should enter into commerce: the second is, that you should take time to consider well in what direction you will employ your capital and energy—time to travel, if you are inclined, before deciding. Then, when you have decided, he will find whatever capital you may require beyond that already at your command. But there is to be no deed of partnership. You are to be prepared to take the full responsibility of your own transactions.'

Philip was silent. It required time for the mind to grasp the full meaning of this proposal. That it was a magnificent one, he felt; indeed it was the magnificence of it which perplexed him. He was to be hoisted at once into a prominent position in the commercial world, although he was without experience of business, and was not conscious of possessing any special aptitude for it. His father knew him better than his uncle did, and had declared him unfitted for commercial pursuits.

He mentioned these objections to his uncle's plan; but the lawyers only smiled at the idea of a man even thinking of such disqualifications as obstacles to his own immediate gain.

'I have known many men who were slow enough to give away a fortune,' said Mr Hawkins, emphasising his words by rubbing his bald head with the eye-glasses, as he gazed almost reproachfully at this singular young man; 'but I never before met a person who was slow to accept one.'

'I daresay; but this position is a little curious. You may set aside at once the project that I should take the money and do nothing for it. Mr Shield's wish is sufficient to bind me to go into trade of some sort; but in doing so, I may make ducks and drakes of his gift in no time.'

'My dear sir, money always makes money if it be guided with even moderate prudence; and I give you credit for possessing that quality to a sufficient degree.'

Philip bowed in acknowledgment of this good opinion.

'Besides, Mr Shield does not mean that you should be set adrift without rudder or compass. He will be always ready to advise; and I need not say

that you may always command our best attention. Also he would expect you to appoint some competent person as your manager, who would be capable of directing the course of your affairs.'

'Ah—Wrentham would be the man, if we could only make it worth his while to join me.'

'We have no doubt, from what we know of Mr Wrentham, that he would consider it much to his advantage to undertake any charge with which you may be disposed to intrust him.'

'I must have time to think over it all,' said Philip, whilst a thousand visions were dancing before his mind's eye, like the dazzling spray of sparks struck from iron at white-heat by a blacksmith's sledge-hammer.

'Certainly, certainly. It is especially mentioned that you are to take whatever time you may require to settle how you shall proceed. Mr Shield is anxious to see you begin operations, but he has no desire to hurry you.'

'I will write to him as soon as I see my way. I suppose this is all you have to tell me?'

'There is only one other trifling matter. I hope we have made you clearly understand that Mr Shield does not insist upon anything. He merely expresses a wish.'

'And I have told you how I regard his wishes—as fixed conditions of my being thought worthy of all this generosity.'

'He is emphatic in desiring that you shall not regard them as conditions, but as mere indications of what he would be most pleased to see you do.'

'Well, what is the remaining wish or condition? It is all the same what we call it.'

'It is, that in the event of your entering into business, he would like you to remember how much more freely and independently a man may act when unshackled by domestic ties. In short, he would like you to remain a bachelor for the first two or three years, until you have firmly established your position.'

'Oh!' exclaimed Philip in a soft crescendo scale; whilst Mr Jackson nodded and grinned, as if there were a good joke somewhere. 'I cannot promise

that.'

'No promise is required; and Mr Shield would not consider it binding if you made it.'

'I am not likely to make it,' was the reply, with a hesitating laugh; 'but this may seriously affect my decision.'

Mr Hawkins was unable to conceive any possible decision except one, and was again gravely effusive in his congratulations. Mr Jackson, shaking hands with Philip at the door, expressed his unqualified approval of the whole scheme in one short phrase: 'You *are* a lucky dog.'

Philip was not sure whether he was a lucky dog or not. His uncle's proposal was liberal and generous beyond all expectation; but there was something—he did not know what—about it that was perplexing. Probably, it was the fact that for the first time he was brought face to face with the necessity of deciding promptly in what course his whole future was to be directed. Hitherto there had been no hurry; and at the time when thoughts of Madge had brought him to serious consideration of how he could most rapidly win a position for her, the invitation from his uncle had arrived. The final decision was again postponed, as it was his duty to obey that call for his mother's sake.

Now his future had been decided for him; and the prospect was in every way a tempting one. There would have been no hesitation on his part, but for the strange position which his father and Mr Shield occupied towards each other. The question Philip had first to settle with himself was, how he should act in order to bring about a reconciliation between them. He knew that if he could accomplish this, he would fulfil his mother's dearest wish—an object nearer his heart than even the possession of a fortune.

As for business, although he had no special inclination for it, he did not dislike it. He had heard and read of millionaires—their struggles and victories, as desperate and as glorious as any recorded in the history of battlefields. Life and honour were as much at stake in doing the daily work of the world as in shooting down the foes of the nation or the foes of the nation's policy. Our merchants, our inventors, our educators, our labourers, were the true soldiers, and their victories were the enduring ones. There was the great enemy of mankind, Poverty, with his attendant demons Ignorance

and Laziness, still to conquer; and there were legions of starving people crying out to be led against him. Vast territories lay untilled, vast resources of earth, air, and water still unused, to be called forth to content and enrich the hungry and poor. What noble work there was for men to do who had sufficient capital at command!

He had never before speculated upon such a career. Now that it was presented to him, his imagination was stirred by thoughts of the great deeds that were yet to be done to bless humanity and ennoble life.

(To be continued.)

SUAKIM.

THE intense interest with which all eyes have been turned upon the Soudan—that is, Country of the Blacks, or Negroland—gives a special value now to any information about that region, particularly if it refer to such towns as Khartoum, or that named at the head of this paper. The former place has been pretty fully described of late in the newspapers, while little has been told us of the latter beyond actual war-news. This is the greater pity, as Suakim possesses a good deal of historical interest, and Khartoum does not.

Suakim—the word is spelt in a variety of ways—is not only one of the most important towns of Nubia, but the chief port of the Soudan and of the whole western coast of the Red Sea. It came into the possession of Egypt in 1865 by cession or purchase from Turkey—along with Massowah and one or two other towns and the districts around them—and now appears to be regarded by the British government and every one else as an integral part of the Egyptian dominions. Similar subjection of Suakim to Egypt, as we shall presently see, existed in very remote times. The town proper lies on a small island about eight miles and three-quarters in diameter—almost as long as the little bay in which it is placed, a mere tongue of water separating it from the mainland.

Crossing the inlet southwards to the mainland, we step into the large suburb called El Gêf, with a much larger population than the insular town, very irregular streets, and the houses mere native (Bishareen) huts. There is also a very lively bazaar, and, in the north-west of the place, the barracks, one section of which, a few years ago, was armed with three pieces of cannon. In the outskirts are the wells—surrounded by gardens and date plantations—which supply the people with drinking-water, although, from the nearness of the wells to the sea, this is brackish, and would scarcely be considered palatable by foreign troops. El Gêf is really an oasis; all round it, save seawards, extend many miles of salt and arid wilderness. Indeed, the whole distance from Suakim to Berber—two hundred and eighty miles inland—is for the most part desert, the route garnished here and there with wells of water and encampments of the wandering Bishareen, who, with the

Haddendowa, a similar set of people, possess the whole wilderness from east of the first cataract of the Nile up to Kassala and the boundaries of Abyssinia. These tribes, though sometimes called Bedouin, whom in many respects they resemble, are really a very different people. Bedouin proper are Arabs of the Semitic, while the Bishareen are of the Hamitic family.

The chief articles of export are cotton, gum-arabic, cattle, hides, butter, tamarinds, senna leaves, and ivory. The imports consist of cotton goods, iron, wood, carpets, weapons, steel, and fancy wares. Berber in the east, and Kassala in the south, are the great centres for all the caravan traffic of Suakim, which is also the port on the one side for the whole Soudan—an inland country as large as India—and on the other side, for Arabia. Hence it is much visited by Mohammedan pilgrims to Mecca, their port of Jeddah occupying a corresponding position on the Arabian to that which Suakim does on the African coast. Twenty years ago, from three to four thousand slaves per annum were shipped from here to Jeddah, and though this monstrous traffic has been much crippled of late years by the Egyptian government, out of regard for English feeling, it is to be feared that it is not yet extinct. Oddly enough, Hassan Mousa Akad, one of the ringleaders in Arabi's recent rebellion, and the greatest slave-merchant in Egypt, was exiled to this very slave-port of Suakim, hence his complicity in the Soudan disturbances is not unnaturally suspected. The total population of the town and suburb is estimated by Schweinfurth—one of our greatest authorities—at from eleven to thirteen thousand. The port is now in regular communication with Suez by steamer—four days' journey—and with Europe by telegraph. The Egyptian governor (Mudeer) and vice-governor (Wakeel) live at Suakim, and the budget for the district in 1882 was—income, £25,945; expenditure, £20,492—thus being one of the few districts of the Soudan which yielded a surplus.

In ancient times, the whole of what we may call the Suakim seaboard—extending northwards along the coast as far as a line drawn from the first cataract, and southwards as far even as Bab-el-Mandeb—was known as the Troglodyte country. The Troglodytes, as the name implies, dwelt in caves, were by occupation herdsmen, and often uncivilised and wretched in the extreme. A graphic picture of the hard life of another Troglodyte people, dwelling in the rocky fastnesses east of Jordan, is preserved for us in the thirtieth chapter of the book of Job. 'For want and famine,' it says, 'they are

solitary; fleeing into the wilderness in former time desolate and waste. Who cut up mallows by the bushes, and juniper roots for their meat. They were driven forth of men (who cried after them as after a thief), to dwell in the cliffs of the valleys, in caves of the earth, and in the rocks.'

Perhaps the Troglodytes of the Nubian shore were a superior stock of their kind; at anyrate, they appear to have been impressed into the army of the ancient Pharaohs, and to have shared in the first invasion of the kingdom of Judah, and the first spoliation of Solomon's Temple. The name of the Pharaoh of that time was Shishak, and two accounts of his expedition have come down to us: one is in the historical books of Scripture (2 Chronicles, xii., also 1 Kings, xiv.); and the other, remarkably enough, is by Shishak himself. That of the Egyptian king is contained in the famous hieroglyphic inscription on the walls of the temple of Karnak at Thebes, in Upper Egypt, a great part of which is still legible, after the lapse of nearly three thousand years! The book of Chronicles tells us with what an immense army of charioteers, cavalry, and infantry, Shishak overran Judea. He marched against it 'with twelve hundred chariots, and threescore thousand horsemen: and the people were without number that came with him out of Egypt; the Lubims, the Sukkiims, and the Ethiopians.' Of these three allies, the first are probably the Libyans (as in Daniel, xi. 43), and the last the same as the modern Abyssinians. For the middle name of 'Sukkiims,' the old Greek translation of the Bible—made by Jews a century or two before the birth of Christ—substitutes the word *Troglodytes*, the very people of the Nubian coast whom we have been considering, and who are now known as Bishareen. But yet more, Pliny the elder, an old Latin writer, who died A.D. 79, mentions, in his enumeration of places on this Troglodyte coast, a town called Suche, which, according to the general opinion of scholars, is identical with the modern port of Suakim, at present (while we write) governed by an English admiral, and its fortifications manned by British sailors and marines.

MISS MARRABLE'S ELOPEMENT.

IN TWO CHAPTERS—CHAPTER II.

MISS MARRABLE, who, when she received this love-letter, was sitting in her bedroom, was thunderstruck. At first, she thought of going to Amy and charging her with baseness and ingratitude; but after some reflection, she decided to let matters, for the time at least, take their course, and to confound the schemes of the rash couple by means of a grand stroke at the final moment. She went, however, at once to Lucy, in whom, as I have said, she had great confidence, and told her all.

‘How foolish of her,’ said Lucy.

‘Yes, my dear! how foolish, and how wicked!’ assented Miss Marrable. ‘I feel it my duty to prevent the carrying out of this mad plan, and also to make Amy suffer for her folly. I shall therefore send her this letter; and allow the hare-brained pair to mature their schemes.—And what, Lucy dear, do you think that I propose to do? You will never guess. Listen! Amy and I are of much the same height. I shall personate her by concealing—ahem—my face, and drive away with this vile young man; and then, when he believes that he has left me far behind, I shall overwhelm him with shame and confusion.’

Lucy could not help laughing. ‘That would really be good fun, aunt,’ she said. ‘Yes, send the letter to Amy; and by all means let matters take their course for the present.’

Miss Marrable did send the letter; and Amy duly received it, unsuspectingly; but five minutes later, Lucy revealed the whole plot to her, and threw her into the deepest trepidation.

Here, however, Lucy’s superior coolness came in most usefully. ‘You need not despair,’ said the elder cousin. ‘If aunt thinks of having fun with you and Mr Jellicoe, why not turn the tables, and have fun with her? You must find some other way of carrying on your correspondence; but at the same

time answer this letter by the old medium. Your answer will of course fall into aunt's hands. You must mislead her, and then'——

'But,' objected Amy, 'how am I to make matters turn out properly?'

'Listen!' said Lucy. 'Aunt proposes to personate you. Very well. Put off the time of your elopement, say, for half an hour; and meantime Mr Jellicoe must find some one to personate him. My idea is for aunt to elope with the billiard-marker, and so give you time to get away. Do you see?'

Amy could not at first grasp the significance of this bold proposition; but when she succeeded in doing so, she was delighted with it.

'I shall tell Mr Rhodes,' said Lucy, when she had sufficiently explained the plan; 'for I know that he will gladly help you; and Mr Jellicoe can talk it all over with him and have the benefit of his advice.'

'But what will aunt say when she discovers how we—how you—have deceived her?' asked Amy.

'Ah!' said Lucy slyly, 'I must talk about that too with Mr Rhodes. But never fear!' And she went off to rejoin Miss Marrable, who was still much flurried.

Later in the day, Lucy met Robert on the beach, and told him what had happened. 'And now,' she said in conclusion, 'I am going to make a dreadful proposition to you. We must also elope together!'

'I am sure I don't mind,' said Mr Rhodes. 'After hearing your news, I was going to propose as much myself. It would take you out of the reach of your aunt's reproaches, when she finds out the trick that has been played upon her.'

'You are a dear old love!' cried Lucy with enthusiasm. 'I wouldn't for the world have Amy made unhappy; and I feel that I must help her, although I don't approve of elopements. Now go and talk to Mr Jellicoe; and don't forget to have the licenses ready. Perhaps Mr Jellicoe can arrange for both Amy and me to sleep that night with the Joneses, whoever they may be; or perhaps, after all, we had better not go there, since aunt knows of that part of the scheme.'

'I daresay,' said Robert, 'that I can arrange for both of you to sleep at the Browns at Llanyltid. They have a large house, and, curiously enough, my

sister Dora, whom you have often met in town, is staying there with them; so you will have a companion and sympathiser. And now I will go and talk to Jellicoe.'

I need not follow in detail the progress of the new scheme of double elopement. Suffice it to say that the bogus correspondence destined to mislead Miss Marrable, was steadily kept up; that Amy and Vivian found other means of safely communicating with one another; that the Browns were written to; that the licenses were obtained; that three carriages-and-pairs were engaged, one to call at the hotel at nine o'clock P.M., and two at half-past; that coachmen were liberally feed; and finally, that the billiard-marker at the *Cors-y-Gedol*, a spruce young fellow of some education, was bribed, at considerable cost, to personate Vivian Jellicoe and to run away with Miss Marrable.

At length, Wednesday morning arrived; and with it came the last of the billet-doux that were to fall into the cunning spinster's hands. One of them had been composed by Vivian and Robert, and written by the former on pink paper, folded billet-doux-wise. It ran as follows:

MY OWN AMY—I have satisfactorily arranged everything. The carriage will be at the door of the hotel at nine o'clock. I shall not show myself, for your aunt may be about. Be careful, therefore, to avoid her; and enter the carriage as quickly as possible. In order that there may be no mistake, I have told the driver to wear a white choker round his neck. I hope that you will be punctual. Everything depends upon punctuality. Till nine o'clock, good-bye.—Your most devoted

VIVIAN.

Miss Marrable, after reading this note, refolded it as usual, and took care that it reached Amy. Then, with the consciousness that she was about to perpetrate a great and good action, she sat down in her own room, and waited for Amy's reply to be brought to her by the treacherous maid. The note, which was very brief, came to Miss Marrable in less than half an hour. 'DEAR VIV,' wrote Amy, 'I will be ready, and will look out for the white choker.—Your loving A.'

In spite of the ordeal which was before her, the good old spinster was perfectly calm and unflurried. At one o'clock she made a very hearty luncheon; at half-past two she took her nieces for a walk, and talked to them with extraordinary affability about the emancipation of women; and at half-past six she appeared at the *table d'hôte*, and, just as if the occasion were an ordinary one, complained of the soup being too peppery, the fish too cold, and the mutton too underdone. Her coolness was admirable. Lucy and Amy, on the other hand, could scarcely conceal their excitement and agitation. They each looked at least a hundred times during dinner at the clock upon the mantel-piece; and they each started and turned red whenever the noise of carriage-wheels without was heard. After dinner, Miss Marrable went again to her room and began to make her preparations.

'How sad it will be,' she thought to herself, 'for poor young Jellicoe when I discover myself and overwhelm him with reproaches. Men are but poor creatures. Perhaps he will faint. Yes; I will take my salts-bottle.' She wrapped herself in an ulster belonging to Amy, and having shrouded her face in a thick veil, took a seat at her window, which happened to be immediately above the front-door of the hotel.

Meantime, Edward Griffiths the billiard-marker was ill at ease. He knew Miss Marrable by sight, and looked forward with terror to the prospect of an encounter with her at close quarters. Nevertheless, he had Vivian Jellicoe's five-pound note in his pocket, and he was determined to see the affair bravely through. He felt, however, that his natural bravery would not be sufficient to support him; and he therefore, at about six o'clock, began to swallow a succession of potent doses of whisky-and-water, with the object of laying in a stock of Dutch courage. Whether the whisky was bad or the water was too powerful, I cannot say; but at ten minutes to nine, when Vivian Jellicoe arrived to give final directions and counsel to his substitute, he found Edward Griffiths decidedly the worse for liquor. Fortunately the young fellow was neither quarrelsome nor noisy in his cups. His main ambition seemed to be to go to sleep in peace; and no sooner had Vivian bundled him into one corner of the carriage, which was in waiting in the stable-yard, than Mr Griffiths incontinently slumbered. The carriage was then driven round to the front-door of the hotel. Miss Marrable, from her post of vantage, saw it, and, remarking that the coachman wore a white choker, descended at once, and listened, as she went, outside Amy's room,

to satisfy herself that that young lady had not forestalled her. The porter with alacrity opened the carriage-door. In the dark shadows of the interior, Miss Marrable caught sight of the figure of a man; and making sure that all was right, she entered at once. An instant later she was being whirled northward along the lonely Harlech Road.

Half an hour afterwards, two other carriages left the hotel, but in the opposite direction. In one of them were Lucy and Mr Rhodes; and in the other, Amy and Mr Jellicoe. It was nearly midnight ere they arrived at the Browns' house at Llanyltid; but the Browns were all up and waiting for them, and the two runaway couples were warmly welcomed, and hospitably taken care of.

Miss Marrable was less fortunate. As soon as the carriage in which she sat had been driven beyond the lights of the town, she threw aside her veil, and gazed with magnificent scorn towards the dim form upon the seat in front of her. The look eliciting no response of any kind, Miss Marrable ventured to cough, at first gently, and then with considerable violence; but still the figure took no notice.

'This is exceedingly strange,' thought the spinster lady. 'I must adopt more active measures.' And with great tenderness, she prodded Mr Griffiths with the point of her umbrella. The billiard-marker groaned in his sleep. 'Mr Jellicoe!' she exclaimed in her deepest and most threatening tones. She had counted upon this exclamation producing an instantaneous and astonishing effect upon her companion; and she was wofully disappointed when he merely groaned again.

'Gracious!' she said to herself: 'he is ill. He would never go on like that, if he were not ill. The fright has been too much for him. Oh, how sorry I am! These men are such weak creatures. I must stop the carriage!' And, throwing down the sash of the window, she put out her head and cried to the driver to pull up his horses. But the driver, like the billiard-marker, had been very liberally feed; and he was determined that nothing should stop him until he reached Harlech; he therefore cracked his whip, to drown Miss Marrable's voice, and drove down the next hill at a pace which threatened to shake the carriage to pieces.

'Stop, stop! For goodness' sake, stop!' shouted Miss Marrable; but finding that her words were not listened to, she drew in her head, and strove to

revive the wretched man in front of her. She held her salts-bottle to his nose; she chafed his hands; she fanned his brow; and she allowed his feverish head to rest upon her shoulder; but she could not awaken him.

‘If he should die!’ she thought. ‘I intended to frighten him; but not so much as this. Oh! this is terrible!’ And once more she tried to prevail upon the driver to stop; but in vain. The sight of distant lights, however, gave her at length some satisfaction. The carriage entered a long avenue, the gate of which lay ready opened for it; and about an hour and a quarter after leaving Abermaw, it drew up before the Joneses’ house near Harlech.

With a sigh of relief, Miss Marrable threw open the door and sprang out, to find herself in the presence of half-a-dozen people who were congregated upon the steps.

‘Quick!’ she cried; ‘don’t ask questions! He is ill; he is dying. Take him out!’

The Joneses, who had not been prepared for the apparition of a middle-aged spinster, and who were expecting Mr Jellicoe and Miss Allerton, were somewhat astonished.

‘Who is inside?’ asked Mr Tom Jones, the son and heir of the family.

‘Oh! Mr Jellicoe! Be quick! For mercy’s sake, be quick!’

‘You don’t mean it!’ cried Tom, rushing to the carriage to succour his friend. But an instant later he burst into a violent fit of laughter. ‘Why, it’s not Jellicoe at all!’ he said. ‘It’s Griffiths, the billiard-marker from the *Cors-y-Gedol*; and he is hopelessly drunk. Nice companion, indeed!’

Miss Marrable is, as I have already said, a woman without weaknesses. On hearing this announcement, however, she fainted away. When, thanks to the kind attentions of the female members of the Joneses’ family, she revived, she indignantly charged those estimable people with having deliberately plotted her discomfiture; and she insisted upon at once returning to Abermaw; but the carriage (and Griffiths) had gone; so Mr Jones, senior, who grasped the situation, volunteered to drive Miss Marrable back to the *Cors-y-Gedol Hotel*; and by twelve o’clock, or shortly afterwards, she was again in her own room. It was then that she learned of the desertion of Lucy and Amy. I need not describe how she received the news, and how she declared that her abandoned nieces should never again behold her face; nor

that, although she is a woman without weaknesses, she passed the greater part of the remainder of the night in violent hysterics. She telegraphed next day to Mr Larkspur and Mr Allerton; and repairing to the *Red Cow*, furiously denounced Sir Thomas Jellicoe as the basest and most heartless of men!

Three weeks afterwards, however, the edge of her anger had worn off. Lucy and Amy were married. It was foolish, but, perhaps, it was not wholly inexcusable; and thus reasoning, Miss Marrable, in the goodness of her heart, determined to gradually receive them back into her favour. But she has never wholly forgiven Lucy for suggesting the substitution of the billiard-marker for Vivian Jellicoe.

‘My dear,’ she says, when she retells the story of her drive to Harlech, ‘the wretched man was perfectly saturated with whisky, and I really don’t know what he might not have done if I hadn’t kept my eye steadily on him. But beneath my gaze he cowered, my dear, positively cowered! I never saw a savage brute so completely tamed.’

And to this day Miss Marrable believes that but for her Eye, the billiard-marker might—horrid thought!—have run away with her too.

A CURIOSITY IN JOURNALISM.

IN the case of such a curiosity in official journalism as the *Police Gazette*, formerly known as the *Hue and Cry*, the public will be interested to learn a little more than the newspapers have briefly announced about the changes made in it by government authority. The paper itself, which was commenced shortly after the formation of the metropolitan police force in 1828, is not allowed to circulate beyond constabulary circles; but its efficiency of management unquestionably concerns the general community. Previous to the year 1828, the metropolis, like other centres of population, was under the care of the old parochial Watch, who, as corrupt as they were feeble, became an absolute street nuisance. Far from being a terror to evil-doers, their notorious negligence and inefficiency enabled the midnight burglar or daring footpad to pursue his criminal avocation with comparative impunity. Peel's Act introduced a greatly improved *régime*; and the new police, nicknamed after their originator, were for a long time popularly known as 'Peelers.' The newly established force required new methods of working, and one of these was the starting of an official newspaper which, though it is perhaps the only one the public never see, has nevertheless often done them good service, and is now to be made of still more value.

It is probably known to few that there exists in connection with the Detective department at Scotland Yard a regular printing establishment, from which sheets are issued four times a day containing information as to persons 'wanted,' current offences, property stolen, lost, or found. A daily list of property stolen is also printed, and distributed to all licensed pawnbrokers. Particulars received from country constabulary forces are inserted in these publications, which are carefully read at parades and studied by the detectives. This, however, only applies to the metropolis; and a strong desire has long prevailed at headquarters to make that larger medium of publicity, the *Police Gazette*, more useful as a means of intercommunication between the whole of the two hundred and ninety police forces of the kingdom. Until the beginning of the present year, that wretched print had shown scarcely any progress or improvement since it was commenced. Its direction has hitherto been nominally in the hands of

the chief clerk at Bow Street police court. In the past, much of its space has been wasted by the frequent repetition of details as to trifling cases; and no systematic arrangements were made for the widespread circulation of the paper among those for whom it is specially intended. The editorship has now been committed to Mr Howard Vincent, director of criminal investigations, who will be assisted by Chief-inspector Cutbush of the executive department at Scotland Yard. It is to the initiation of Mr Vincent that the improvements now made are chiefly due; and it may be remembered that in his presidential address to the Repression of Crime Section of the recent Social Congress at Huddersfield, that gentleman explained his intentions. The proposals he made were so favourably received, that subscriptions amounting to nearly one thousand pounds were placed at his disposal. These, however, have not been needed, as it happens that the improvements have been accompanied by an actual reduction of expense; and the Home Secretary has determined that the costs, limited within a certain moderate sum, shall still be borne entirely by the public funds.

In addition to being much better printed, the new *Gazette* already shows decided improvement both in the selection and arrangement of its contents. For convenient reference, particulars are not only grouped according to the usual categories of crime, but are now classified under special headings for the various districts to which cases belong. Illustrations have also been introduced as a new feature. These take the form of woodcuts from photographs of persons 'wanted' on various charges, or of valuable articles stolen. The first number of the *Gazette* contains the likeness of several criminals of whom the authorities are in pursuit. In one instance, so as to aid identification, the subject is shown not only with beard and moustache, but also as he would appear when clean shaved. Some of these faces, it is true, seem decent and commonplace enough, such as one sees almost every hour of the day in the public streets; but others, 'an index of all villainy,' are unmistakably those of dangerous characters whom none of us would like to meet alone in a quiet road on a dark night. But it is in the police album^[1] that we can best study the variety of expression by which the human countenance can betray every shade of criminal depravity.

Meantime our business is only with the *Gazette*, which, among other changes, has altered its days of publication. Hitherto it has been issued three

times every week; but now that the space is more carefully utilised, twice a week is found sufficient. The War Office and Admiralty have always had the privilege of inserting in its pages a list and description of deserters from the army and navy. In future, the Tuesday's issue will be entirely devoted to these matters; and when it is known that last year the total number of deserters was only one short of six thousand, it may be inferred that the weekly list does not leave much space to spare in a small four-page paper. The Friday's issue extends to eight pages, and is reserved exclusively for police information, with the exception of two pages now set apart by contract for advertising purposes. As far as increased circulation is concerned, arrangements have been made to send supplies of the *Gazette* not only to every police force in the United Kingdom, but also, through the government offices, to the guardians of the peace in the British colonies and India. From the public generally, the *Gazette* is withheld.

The early issues of the *Gazette*, especially between 1829 and 1831, bear significant testimony to the labour disturbances and political excitement which immediately preceded the passing of the great Reform Bill. Every number was then largely occupied with royal proclamations in the cause of order, and offers in Lord Melbourne's name of government rewards for the arrest of incendiaries and disturbers of the public peace. Again we are on the eve of parliamentary reform, but without any symptoms of rioting; and the improved columns of the *Hue and Cry* are now left more free for ordinary police information as to the appearance and lawless doings of the 'incurable' class.

THE MONTH: SCIENCE AND ARTS.

FROM the Report issued by the Committee appointed to consider the best way of rebuilding the houses at Casamicciola destroyed in the recent earthquake, we learn that that terrible catastrophe occasioned the deaths of no fewer than two thousand three hundred and thirteen persons, and injury to seven hundred and sixty-two more. Although these unfortunates did not all actually belong to the island, there were among them only fifty-four who could be called foreigners. It will probably be found advisable to rebuild the ruined habitations on the pattern adopted in certain places of Central America, where earthquakes are common. The houses there are built of such light materials, that when a shock comes, they rattle down like a veritable house of cards, and can almost be rattled together again as easily when the danger has for the time passed. In London and some other of our cities and towns, the houses are so shamefully run up that a very mild shock of earthquake would suffice to shake them to pieces.

We are apt to look upon these jerry-built houses as the result of competition and the continual cry for cheap houses. On the other hand, we regard our cathedrals as solid monuments to the more honest work of former times. But this notion must be dispelled. The Peterborough Cathedral architect has been examining the foundations and piers of the tower of that fabric, which it will be remembered he some time ago reported to be in a dangerous condition, and they turn out to be as perfect an example of jerry-building as could be found in our own enlightened times. The piers were found to consist of a thin facing of stone, the interior being filled in with small rubble-stone and sandy earth. He tells us that 'it is impossible to conceive a worse piece of construction, and it is equally impossible to understand how it is that these piers have stood so long.' The piers have simply been enabled to hold together by the strength of their exterior clothing. It is some small satisfaction to the modern householder that dishonest building has not been invented for his especial torment, but was practised as long ago as the fourteenth century.

Another far more valuable relic of the past is, as we recently indicated, exciting attention on account of its decaying condition. Westminster Abbey, which may justly be regarded as the most important ecclesiastical building in the kingdom, is wasting away piecemeal under the effects of London smoke and atmospheric agencies generally. The sum required for its restoration is estimated at eighty thousand pounds, and this is probably short of the real amount which will be required to do the work effectually. For such a national purpose, the purse of the nation ought undoubtedly to be responsible.

The complete Report of Professor Hull's labours, as chief of the little band of scientific explorers who have just returned from a geological survey of Palestine, will be looked forward to with unusual interest, for he brings back with him materials for constructing a far more complete map than has ever before been possible. The ancient sea-margins of the Gulfs of Suez and Akabah have been traced at a height of two hundred feet above their present surfaces—indicating that the Mediterranean and Red Seas have been at one time in natural connection with one another. Professor Hull believes that this was the case at the time of the Exodus. The terraces of the Jordan have also been examined, the most important of these ancient margins being six hundred feet above the present level of the Dead Sea. Besides his scientific Report, the learned Professor is preparing a popular account of his pilgrimage, which will duly appear in the Transactions of the Geological Society. His journeyings will cover much of the same ground traversed nearly fifty years ago by David Roberts, whose drawings of the places visited aroused so much interest at the time, and which have never since been surpassed.

Not very many years ago, a map of Africa presented in its centre a blank space, which was explained to inquiring children as indicating a country so hot that nobody had been there or could live there. This benighted region has now an atlas all to itself. Under the auspices of the Geographical Society, Mr Ravenstein has just completed their map of Eastern Equatorial Africa; it is of large size, and contains altogether twenty-five sheets. He will now commence a similar work for Western Africa, and has proceeded to Portugal in order to take advantage of the materials in the possession of that government bearing upon the subject. This work is also undertaken for and at the expense of the Geographical Society.

The official Report of the late census in British Burmah is not without interest to dwellers in Britain. Only two languages had to be used in the process of enumeration—namely, Burmese and English. The people at first thought that the strange proceedings heralded the advent of a new tax, and one tribe fled across the frontier so as to be out of the way. Another idea that occurred to the people was that the English made use of human heads for inquiring into the future. But these difficulties having been smoothed over, the census was taken satisfactorily. British Burmah is, roughly speaking, of the same area as Great Britain and Ireland, with a population less than that of London. This population, under British rule, has doubled in twenty years, and there is every sign of its continued increase. The males are far in excess of the females, and what seems a very important key to the wonderful prosperity of the country is the fact that there are ten acres of cultivated land for every eight persons living in it.

It is reported that Baron Nordenskiöld, whose recent explorations in and around Greenland aroused so much interest in scientific circles, is contemplating a voyage next year to the south polar regions. The cost of the projected expedition is nearly two hundred thousand pounds, but this seemingly large sum will include the expense of building a ship of special construction, to meet the requirements of the explorers.

International courtesies are so very few and far between, that when one occurs it is worthy of the most honourable mention. Many years ago, a band of English Arctic explorers abandoned their ship, the *Resolute*, for it was hopelessly frozen into the ice-pack. The ship, however, at last floated free, and was taken by an American whaler to New York. The gallant Americans thereupon put the vessel into splendid order, and presented her to Queen Victoria. It was but the other day that the old ship was broken up, when a desk was made from her timbers and presented to the American President. The British government have now presented the *Alert*, which has also seen Arctic service, to the United States government for the use of the Greeley relief expedition. The ship has long ago been strengthened with teak for protection against the ice, and is thus well fitted for the purpose in view.

The recent experiments at Folkestone once more proved the value of throwing oil on troubled waters, the efficacy of which operation in stormy weather we described last month. In addition to the oil-shell there mentioned, another invention falls to be noticed, by which the same gun

from which the oil-shell is discharged may be also employed for projecting a heavy solid cylindrical shot, to which is attached a flexible tubing. Upon firing the gun, the shot is carried a long distance out to sea, pulling the tube after it. The shot sinks to the bottom, and the tube thus anchored can be used with a pump for forcing the oil to any spot in the neighbourhood. This contrivance, like that of the oil-shell, is the invention of Mr Gordon.

The preparations for the International Health Exhibition to be opened in London in May next, proceed very rapidly. The eight water-companies which supply London, and which just now are being so roundly abused on the score of overcharges, will exhibit the various apparatus employed by them for the supply, filtration, &c., of water. They will also combine in erecting an immense fountain in the grounds, the jets of which will be brilliantly illuminated at night by electricity.

An American paper gives an interesting account of the manufacture of 'Yankee sardines,' which may be explained to the uninitiated to mean small herrings preserved in oil and flavoured with spices, to imitate the sardines of French preparation. To begin with, the fish are laid in heaps on long tables, where they are rapidly cleaned and decapitated by children. The herrings are then pickled for one hour, to remove a certain tell-tale flavour which they possess, after which they are dried. The next operation is to thoroughly cook them in boiling oil; and finally, they are packed in the familiar square tins, and duly furnished with a French label, such as, 'Sardines à la Francaise,' or, 'A l'huile d'olive.' The free, or rather the true translation of this latter inscription would be, 'cotton-seed oil,' and, sad to say, not always of the first quality.

A paper dealing with an outbreak in a German town of that terrible disease known as trichinosis was recently read before the French Academy of Medicine. It is worthy of attention as going far to prove that this disease, usually contracted by the consumption of unwholesome pork, is avoidable, if the ordinary precaution of thoroughly cooking the food be resorted to. In the case in question, more than three hundred persons were attacked with the disease, and of these nearly one-sixth died. It was proved beyond question that all the victims ate the meat absolutely raw, it being the custom to chop it fine and to spread it like butter on slices of bread. One single family, which consumed some of the same meat in the form of cooked sausages, exhibited no trace of the disease. It may be mentioned that a

certain dose of alcohol exercised a most favourable effect in diminishing the virulence of the complaint.

A new system of railway signals which is worked by electricity, instead of by mechanical leverage, has lately been experimented upon with great success, but like most other things of an electrical kind, its ready adoption must depend upon its expense as compared with that of the older-fashioned plant. Hitherto, the ordinary electric magnet has been found unequal to this class of work, principally because its power of attraction is only great when very near the object to be attracted, and also because its impact on its armature is so violent as to lead to risk of deranging the apparatus employed. By use of what is known as the long-pull electro-magnet, recently invented by Mr Stanley Currie, these difficulties have been obviated, and signals of every kind can be worked most perfectly through the medium of conducting wires. The system has been at work for the past two months at Gloucester, and is being adopted experimentally in other directions.

The volcano at Krakatoa will long be remembered, if only on account of the wide area over which its products have been distributed. To say nothing of the dust particles which are supposed to have found their origin there, and which are credited with having been the active cause of our late gorgeous sunsets, undoubted volcanic particles have lately been found at Philadelphia. By melting and evaporating the snow upon which these tiny fragments were found, a residue of solid particles was apparent, which the microscope at once pronounced to be of a volcanic nature. It seems difficult to believe that solid matter could thus be carried in the air for four months, during which it must, if it came from Krakatoa, have covered the enormous distance of ten thousand miles. Another supposition is that the volcanic particles found at Philadelphia may have been wafted thither from Alaska, in the north-west corner of North America, where a great eruption has occurred. According to our authority, a submarine volcano shot up there last summer, and has already formed an island in the Behring Sea, from eight hundred to twelve hundred feet high. It is therefore possible that volcanic dust may have found its way, from this source, to the southern states of America, and even to Great Britain. The enormous distances traversed by these glassy particles may be thus accounted for: when steam is forced through a mass of glassy lava, the molten material is shot up with it in the

form of thin filaments, just like spun glass. These, like so many pieces of spider web, would be borne aloft by the air for a very long period.

It seems only yesterday that iron furnace slag was looked upon as a waste product, for which no possible use could be found. It is now made into bricks, into cement, into wool-packing for steam-boilers, and more recently it has been found a most effective material for making all kinds of vases and other things of an ornamental nature. For this purpose, the slag is freed of its coarser particles, mixed with a certain quantity of glass and colouring matter, and when in a molten condition, is stirred about so as to present a veined appearance. It is then moulded into various forms, and is ready for sale.

We lately had an opportunity of visiting the Fine Art Loan Exhibition at Cardiff, which has been opened for three months, for the purpose of collecting funds in aid of the projected Cambrian Academy. The Exhibition includes works by some of our most eminent artists, both living and deceased, as well as a collection of such articles as can be grouped under the head of Art. But a novel feature of the Exhibition is its complete array of telephonic and telegraphic apparatus. By the co-operation of the telegraphic authorities, communication has been opened up by telephone between the Exhibition and Swansea, a distance of fifty-two miles. Not only is speech quite easy over this distance, but the voices of those acquainted with one another are readily recognised. At the time of our visit, the apparatus was connected with the theatre at Swansea, and we had the curious experience of listening to chorus, band, and solo voices, which were rendering a popular opera more than half a hundred miles away.

Mr J. C. Robinson, in the course of an interesting article contributed to the *Times* on the conservation of Sir Joshua Reynolds's pictures, concludes with a recommendation which all owners of valuable oil-paintings should take note of. He strongly advocates the use of glass as a covering for such pictures, and is glad to see that the practice of thus framing them is on the increase. 'This plan,' he says, 'almost entirely obviates the necessity for the periodical rubbing up and cleaning the surface of pictures with the silk handkerchief or cotton-wool, inasmuch as the protecting glass, and not the painted surface of the picture, receives the rapidly accumulating deposit of dust and dirt.' But even this he considers to be only a half-measure. The back of the picture should be stretched over with a damp-resisting sheet of

india-rubber or American cloth, for it requires protection only second to the painted face of the canvas.

In presenting the gold medal of the Royal Astronomical Society to Mr A. Common for his wonderful photographs of celestial objects, the President of that honourable body gave a most interesting history of the medallist's gradual progress in the difficult work in which he so much excels. Mr Common commenced work with a modest reflecting telescope of five and a half inches; but he was not satisfied until he had obtained one measuring no less than three feet across its mirror. He has also turned his earnest attention to the clockwork for driving the instrument, so that as the busy world turns on its axis, the objects focused remain stationary. This is highly necessary, when it is remembered that sometimes a star photograph occupies as much as an hour and a half in the taking, even with the most sensitive plates. This long duration of the action of the feeble light from stars so remote that they cannot be seen by the naked eye, has the effect of impressing the chemical surface so that the invisible is pictured! It is evident that a new field of research is thus opened out; and the President did well in pointing out what great services can be rendered to knowledge by the amateur worker who, like Mr Common, has the means and the ability to employ his time so well.

In these days of oleomargarine, bosch butter, and other mixtures which are supposed to furnish excellent substitutes for the genuine article, it becomes highly necessary to have some means of distinguishing the true from the false. A contribution to microscopical science towards this end is a test discovered by Dr Belfield of Chicago, which will at once identify a fat if it consist either of lard or tallow. Pure lard crystals exhibit thin rhomboidal plates, while those of tallow are quite different, and are of a curved form somewhat resembling the italic letter *f*.

A paper on the Ventilation of Theatres was lately read by Mr Seddon at the Parkes Museum of Hygiene, London. In some crowded theatres, the air has been said by a competent authority to be more foul than that of the street sewers. The intensely heated air would seem to act as a kind of pump, and to extract the vitiated atmosphere from the drains below the building. The successful introduction of the electric incandescent system of lighting to more than one metropolitan theatre has done much to mitigate the evil complained of; but it is quite certain that the ventilation of public buildings generally does not receive the attention which it so imperatively demands.

Another important consideration that is too often neglected is the acoustic properties of public buildings. Even in the last great work which has, after years of labour, been finished in London—we refer to the new law-courts—complaints are constant from those who have to work in them, of the great difficulty both in making their voices heard and in appreciating what is said by others. Public speakers whose duties carry them to various towns and cities throughout the kingdom, know very well that it is the exception, and not the rule, to find a room which is comfortable to speak in. Either the voice falls dead and flat, as if absorbed by a screen of wool; or it reverberates from every wall with such confusing echoes, that the syllables must be uttered with painful deliberation. A Committee appointed by one of our learned Societies to inquire into the reason why some rooms should be acoustically perfect, while others are quite the reverse, would do a vast amount of good. Until such an inquiry is set on foot, architects will continue to design buildings in which this necessary property is quite neglected.

OCCASIONAL NOTES.

INTERESTING NOTES ON TROUT-LIFE.

AT a recent meeting of the Scotch Fisheries Improvement Association, held at Edinburgh, Mr Harvie-Brown communicated some notes on trout-life, which the Association considered of so much scientific interest, that it was resolved to engross them in the minutes. The notes are as follow:

‘The subject of coloration of flesh of trout is a much more intricate one than at first appears. I know of trout holding largely developed spawn in June and July in a loch in Sutherland, whose flesh is not pink only, but bright red like a salmon’s, and yet are not fit to be eaten. I know, also, in a limestone burn the very finest trout, which on the table are perfectly white in the flesh, whatever size they grow to; but in another limestone burn from the same sources, or nearly so, the trout are quite different in appearance externally, but equally white in flesh and equally delicious for eating.

‘I put a quarter-pound trout, along with others, into a previously barren loch. In two years some of these trout attained to four and a quarter pound-weight, developed huge fins and square or rounded tails, lost all spots, took on a coat of dark slime, grew huge teeth, and became *feroces* in that short time. The common burn trout, taken from a very high rocky burn up in the hills, in two years became indistinguishable from *Salmo ferox*. The first year they grew to about a pound, or a pound and a half, took on a bright silvery sheen of scales, were deep and high shouldered, lusty and powerful, more resembling Loch Leven trout than any others. This was when their feeding and condition were at their best; but as food decreased, and the trout rapidly increased in number, spawning in innumerable quantities, and with no enemies, the larger fish began to prey on the smaller, grew big teeth, swam deep, and lost colour, grew large fins and a big head, and became *Salmo ferox* so called. In two years more the food-supply became exhausted; and now the chain of lochs holds nothing but huge, lanky, kelty-looking fish and swarms of diminutive “black nebs,” neither of the sorts

deserving of the angler's notice. The first year they were splendid fish—rich and fat. Now they are dry and tasteless.'

LABOUR AND WAGES IN AUSTRALIA.

It would appear from the latest statistics that during the past few years wages have risen in some trades, and in a few only, have fallen. In the skilled branches of labour especially the tendency has been upwards, and the same thing is also noticeable in agricultural labour. For example, the rates for married couples on stations have risen from fifty-five to sixty-five pounds in 1876 to sixty or eighty pounds in 1883. The wages of farm-labourers have risen to fifty pounds or thereabouts, while only in the case of country blacksmiths have wages declined, the rates for such being now seventy-five to eighty pounds per annum. The colony is stated to be capable of readily absorbing any amount of skilled agricultural labour, especially that of the handy kind, without affecting the current rates of wages. Agricultural labour is in more demand than artisan labour, and good industrious hands would do excellently, as compared with the same class in England, both in regard to food and pay. With regard to other occupations, the following rates are paid on the New South Wales railways: clerks, two hundred to one hundred and fifty pounds per annum; foremen, five pounds ten shillings to three pounds seven shillings per week; draftsmen, four pounds fifteen shillings per week; timekeepers, three to two pounds per week; fitters, 12s. 4d. to 8s. per day; blacksmiths, 12s. 8d. to 10s. 4d. per day; turners, 12s. 2d. to 10s. 2d. per day; pattern makers, 11s. 10d. per day; brass-moulders, 11s. 4d. per day; plumbers, 11s. to 10s. per day; tinsmiths, 11s. to 10s. per day; brass-finishers, 9s. 6d. to 9s. per day; carpenters, 11s. 6d. to 8s. per day; painters, 11s. to 9s. 8d. per day; strikers, 7s. 4d. to 7s. per day; and cleaners, 7s. per day. The working day in the case of many trades does not exceed eight hours.

THE RUSSIAN CROWN ESTATES.

While so much is written of the internal economy of Russia, many will be surprised to hear of the extraordinary extent of the lands which form the estates of the Crown. The extent of the possessions of the Russian emperor may be gathered from the fact that the Altai estates alone cover an area of

over one hundred and seventy thousand square miles, being about three times the size of England and Wales. The Nertchinsk estates, in Eastern Siberia, are estimated at about seventy-six thousand six hundred square miles, or more than twice the size of Scotland and Wales put together. In the Altai estates are situated the gold and silver mines of Barnaul, Paulov, Smijov, and Loktjepp, the copper foundry at Sasoum, and the great iron-works of Gavrilov, in the Salagirov district. The receipts from these enormous estates are in a ridiculously pitiful ratio to their extent. In the year 1882 they amounted to nine hundred and fifty thousand roubles, or a little more than ninety-five thousand pounds; while for 1883 the revenue was estimated at less than half this sum, or about four hundred thousand roubles. The rents, &c., gave a surplus over expense of administration of about a million and a half of roubles, or about one hundred and fifty thousand pounds. On the other hand, the working of the mines showed a deficit of over a million; hence the result just indicated. A partial explanation of this very unsatisfactory state of things is to be found in the situation of the mines, which are generally in places quite destitute of wood, while the smelting-works were naturally located in districts where wood abounds, sometimes as much as three hundred and four hundred miles distant from the mines. The cost of transport of raw materials became considerable in this way. By degrees, all the wood available in the neighbourhood of the smelting-works became used up, and it was necessary to fetch wood from distances of even over one hundred kilometres. Formerly, the mines were really penal settlements, worked by convicts, who were partly helped by immigrants, whose sons were exempted from military service on the condition of working in the mines. But since the abolition of serfdom this system has been quite altered, and there is now a great deal of free labour on the ordinary conditions.

HYDROPHOBIA—IMPORTANT EXPERIMENTS.

M. Pasteur, who has already made so many valuable discoveries in connection with diseases that are propagated by germs, has, in his own name and that of his assistants, MM. Chamberlan and Roux, communicated to the French Academies of Sciences and Medicine the results of his experimental inoculations with the virus of rabies. He finds that the virus may remain in the nervous tissues without manifestation for three weeks,

even during the summer months. Virulence is manifested not merely in the nervous tissues, but in the parotid and sub-lingual glands. The granulations observed in the fourth ventricle, when in a state of virulence, are finer than the granulations in the fourth ventricle when in a healthy state, and they can be coloured by means of aniline derivatives. The virus of rabies injected into the veins or beneath the skin produces paralytic rabies, while inoculations into the spinal cord or the brain produce the paroxysmal form. Inoculations with quantities of the virus too small to be effective, have no preservative influence against subsequent inoculations. Whether the virus is propagated by means of the nervous tissues or by absorption through the surfaces of the wound, has not been ascertained. Finally, the experiments have shown that the protective 'attenuation' of the virus is possible. The energy or the nature of the virus varies in each species of animals. By passing the virus through different animals, 'cultures,' or varying qualities of virus, are obtained, whose precise effects can be predicted. Thus a 'culture' has been obtained which certainly kills a rabbit in five or six days, and another which certainly kills a guinea-pig in the same time. Other things being equal, the virulence varies inversely with the duration of the incubation. M. Pasteur and his assistants have good reason to believe that by means of a special culture they have succeeded in making twenty dogs absolutely proof against rabid inoculations. M. Pasteur, with his usual caution, asks for a little longer time before finally pronouncing on the condition of the dogs in question. To devise a means of making the dog proof against rabies is, of course, to devise a means of almost certainly preserving man (including children) from this frightful disorder; for hydrophobia is almost invariably communicated to man and other animals by the bites of rabid dogs.

THE ELECTRIC LIGHT IN RAILWAY CARRIAGES.

An interesting experiment was commenced just after Christmas last by the District Railway Company, on the short branch line which connects Kensington and Fulham, passing through Earl's Court and Walham Green. On the 2d of January last, the carriages running on this short line were lighted for the first time, each with a small Swan burner, inclosed in a little glass globe; and although only a very small coil of fine wire, thin as a hair, shaped something like a letter U, was employed, the light was so brilliant

and steady that the smallest print could be read by it easily. The experiment lasted about a fortnight or three weeks, and was worked from a luggage van attached to the rear of the train, and fitted up for the purpose. This experiment is interesting, and the result has been most successful, not a slip, nor a hitch of any kind, having occurred; while the reports as to cost are, it is understood, perfectly satisfactory.

Let us hope that this beautiful system of lighting may speedily be introduced on the different railways throughout the country; and especially on the District line of the Metropolitan Railway, where the bad blinking gas is so terribly trying to those who have to make two journeys a day by it, and who desire to employ the time of transit with their book or their paper, which becomes a work of difficulty under the present gas arrangements, but which may possibly be explained by one word, 'economy;' for it is a well-established fact, patent to all, that gas is light and brilliant enough for most purposes, provided a proper and sufficient *quantity* is used.

DISSECTION AFTER DEATH.

Amongst the strange institutions which have been started within the last few years is that of 'The Society for Mutual Autopsy,' which commenced its existence in Paris in the year 1876. No balloting or any elaborate system is necessary to become a member. A proper introduction with a fee of five francs suffices, and an engagement to will your body to the Society for the purpose of dissection after death. In order to prevent the friends and relatives of the dead from frustrating the intentions of the testator, by disposing of the corpse in the usual manner, a proper legal form has been drawn up and inscribed in the Rules. This Society, which consists of about two hundred members, a dozen of whom are ladies, contains amongst its members many men eminent in the medical world in Paris, as well as distinguished in science and art. The theory of the founders is, that in consequence of the difficulty of obtaining for post-mortem examinations any other subjects but those of the lowest classes, whose faculties are naturally warped or otherwise undeveloped, much benefit must accrue to science by an opportunity being given for the dissection of persons of cultivated understanding, and particularly by making observations on the brain. Between twenty and thirty of the members of this Society generally

dine together once a month at a restaurant near the Halles, where they pass a congenial evening, although there is a touch of ghastliness in the gathering. When one of their community is missing at the banquet, instead of lamenting over his departure, every one listens with rapt interest to the surgeon's explanation of the post-mortem examination he has made.

PREVENTION OF BLINDNESS IN INFANCY.

The Manchester and Salford Sanitary Association recently issued a paper, based upon the directions of the Society for the Prevention of Blindness. From it we learn that one of the most frequent causes of blindness is the inflammation of the eyes of new-born babies. Yet this is a disease which can be entirely prevented by cleanliness, and always cured if taken in time. The essential precautions against the disease are: (1) Immediately after the birth of the baby, and before anything else is done, wipe the eyelids and all parts surrounding the eyes with a soft dry linen rag; soon afterwards wash these parts with tepid water before any other part is touched. (2) Avoid exposing the baby to cold air; do not take it into the open air in cold weather; dress the infant warmly, and cover its head, because cold is also one of the causes of this eye-disease. When the disease appears, it is easily and at once recognised by the redness, swelling, and heat of the eyelids, and by the discharge of yellowish white matter from the eye. Immediately on the appearance of these signs, seek the advice of a medical man; but in the meantime, proceed at once to keep the eyes as clean as possible by very frequently cleansing away the discharge. It is the discharge which does the mischief. The cleansing of the eye is best done in this way: (1) Separate the eyelids with the finger and thumb, and wash out the matter by allowing a gentle stream of lukewarm water to run between them from a piece of rag or cotton-wool held two or three inches above the eyes. (2) Then move the eyelids up and down and from side to side in a gentle rubbing way, to bring out the matter from below them; then wipe it or wash it off in the same manner. This cleansing will take three or four minutes, and it is to be repeated regularly every half-hour at first, and later, if there is less discharge, every hour. (3) The saving of the sight depends entirely on the greatest care and attention to cleanliness. Small pieces of clean rag are better than a sponge, as each rag is to be used once only, and then burnt immediately; sponges should never be used, except they are burnt after each

washing. (4) A little washed lard should be smeared along the edges of the eyelids occasionally, to prevent them from sticking. Of all the mistaken practices which ignorance is apt to resort to, none is more ruinous than the use of poultices. Let them be dreaded and shunned as the destroyers of a new-born baby's sight. Tea-leaves and sugar-of-lead lotion are equally conducive to terrible mischief, stopping the way, as they do, to the only right and proper course to be taken.

CARD-TELEGRAMS.

Great as have been recent improvements in our postal service, we have yet to learn something from the Parisians, whose system of Card-telegrams is worthy of notice. The cards are of two kinds—namely, yellow similar to our own, and blue, which, when secrecy is desired, may be closed. By dropping the card into the Card Telegram Box at the nearest telegraph office, it is shot through one of the pneumatic tubes which are now being extended all over Paris, and is delivered at its destination within half an hour. Fifty to seventy words can be written on the card, the cost of which is threepence. It is further intended to permit of cards being dropped into the boxes up to fifteen minutes of the departure of the mail-trains, a boon which merchants in Great Britain may well envy.

HOW AND WHERE THE HERRING SPAWNS.

According to a contemporary, we learn that Professor Cossar Ewart, Edinburgh University, convener of the Scientific Investigation Committee of the Board of Fisheries, was at the beginning of March at the well-known fishing-ground off the coast of Ayrshire known as the banks of Ballantrae, when some interesting investigations were made into the nature of the sea-bottom and spawn deposited on that famous herring-bed. The banks were dredged from a depth of eight to twenty-two fathoms. At a depth of eight to eleven fathoms the bottom was composed of clean gravel, with very little seaweed; beyond the eleven fathoms, clay, mud, and shell. On the stones lifted by the dredge, portions of herring spawn were found firmly attached to the surface of the stones in different stages of development, the more advanced manifesting, in lively action, the embryo herring. Spawn was also taken from the living herring and placed on glasses in hatching-boxes, and

these also showed the eggs in progress of development. From a small stone of a few inches of surface as many eggs were found as, if allowed to arrive at maturity, would have yielded crans of herrings. The information obtained by Professor Cossar Ewart, during his recent dredgings, will be of the greatest importance in throwing light upon a hitherto but imperfectly understood question in natural history.

The banks in the evening presented a scene of lively interest, for as the sun began to set, a school of at least forty whales and porpoises began to play, and, circling around the margin of the fishing-banks, rose and fell in graceful plunges, their black fins and backs rising in curves for a moment, and then disappearing, while the porpoises made wild leaps many feet clear out of the water. Their presence was accounted for next morning, when a good many of the seine trawlers entered Loch Ryan and Girvan with from one to three hundred baskets of herrings each.

Professor Cossar Ewart has since had some more successful dredgings. He has also made some important discoveries regarding natural and artificial spawning, and deposited live herring and a quantity of spawn in the aquarium at Rothesay.

A FLOURISHING FRUIT-FARM.

At Toddington, in Gloucestershire, there has been going on for a few years the cultivation of fruit on a very large scale; a fruit-farm of five hundred acres having been planted by Lord Sudeley, and which, we are glad to know, has proved so successful, that its area is about to be enlarged to the extent of other two hundred acres. An enormous number of fruit-trees of many kinds has been planted, along with thousands of currant-bushes, whilst upwards of a hundred acres of the land are devoted to the growth of strawberries. A noteworthy feature of the scheme consists of a market being found for the smaller fruits on the ground on which they have been grown. In other words, Lord Sudeley has, with great foresight, erected a suite of boiling-houses and packing-rooms, which have been let to an enterprising person, who manufactures genuine jams and jellies from the fruit grown at Toddington. In fruit-preserving, the English and Scotch boilers—and the latter class have largely increased during the last few years—have a great advantage over their brethren of the continent and the United States,

because of the greater cheapness of the sugar, which is required in large quantities. It is to be hoped that the example set by Lord Sudeley will be speedily followed by some of his territorial brethren. As a nation, we could manage to consume much more fruit than we do at present, if we could obtain it at a moderate price. In the orchards at Toddington have been planted as many as thirty-two thousand plum-trees, nine thousand damson trees, and three thousand nine hundred pear and apple trees, while there are no fewer than two hundred and twenty-eight thousand black-currant bushes.

THE GRAPE AND PEACH IN AMERICA.

The old saying about the inutility of carrying coals to Newcastle receives a new rendering in the fact that vine plants are being brought from America to replenish the vineyards of France, which have been in some instances devastated by the phylloxera. Grapes are now extensively grown in the United States both for dessert and wine-making. A lady who has recently been travelling in California, where the grape family is wonderfully numerous, and many of the vines exceptionally prolific, sometimes obtaining a 'luxuriance which sounds almost incredible'—this lady—C. F. Gordon-Cumming—tells us, among other facts, of bunches of grapes which have been found to weigh as high as fifty pounds! The vineyards of Colonel Wilson, in the neighbourhood of the garden-city of Los Angeles, cover two hundred and fifty acres of ground, and the grapes yield one thousand gallons of wine to the acre. In another vineyard, there grow upwards of two hundred varieties of grapes; and in the cellars of its proprietor are stored two hundred thousand gallons of grape-juice, ripening into wine, of which many kinds are made in the state of California. Need it be said that grapes in these regions are cheap—a hatful can be purchased for a few cents! Only think of the above-named Colonel Wilson having 'two and a half million pounds of grapes, hung up by their stalks, to keep them fresh for the market'! That fine fruit, the peach, is equally cheap in the peach-growing districts of the United States. The annual value of the American peach-crop is estimated at eleven and a half million pounds sterling. In some seasons, peaches are so abundant, that, to prevent their being lost, they are used in immense quantities for the feeding of pigs. Cannot this fruit be utilised for consumption in Europe? Supplies of the fresh fruit might be sent to us in the refrigerated chambers of the steamboats.

BOOK GOSSIP.

ONE of the most interesting books of travel issued of late years is that entitled, *Arminius Vambery: His Life and Adventures* (London: T. Fisher Unwin), which is now in the third edition. This Hungarian traveller is a man of rare courage and will, and possessed of high literary accomplishments; and the narrative of his wanderings in various capacities in Asia and Europe is told with a graphic and picturesque power which is extremely captivating.

Vambery, who was born in 1832, had a singularly hard up-bringing, and the story of his early years is quite as interesting as his later adventures in foreign lands. His father died a few months after the birth of the boy, leaving the family in extremely poor circumstances. When he was twelve years of age—up to which time, from lameness, he could only walk with the help of a crutch—his mother thought him old enough to shift for himself. He had previously been three years at school, where he had drawn attention upon himself by his precocity. But the inexorable poverty of his parent stood in the way of further education, and at twelve he was apprenticed to a ladies' dressmaker, but only stayed long enough in this employment to learn to stitch two pieces of muslin together. He left the shop of the 'dress-artist,' and did a little teaching in the family of an innkeeper, 'occasionally waiting on thirsty guests.' When he had saved up eight florins, he hastened from the Island of Schütt, where he had spent his years, to a gymnasium in the vicinity of Pressburg, and here began a strange struggle for existence and education. His money was just sufficient to buy the necessary books, and he had to depend on the kindness and charity of others for his food. Seven different families each gave him one day in the week a free meal, adding to it something for breakfast and luncheon; and he got the cast-off clothes of the wealthier school-boys. Notwithstanding all drawbacks, he made great progress in his studies, and took a high place in the Latin class—he was indeed able at fourteen years of age to speak Latin with considerable fluency. We cannot follow his career further, but can with confidence commend the singular story of his life and adventures to all readers, both young and old.



Literature and angling would seem to have something in common. The number of books that have been written on the 'gentle art,' and that by men of striking ability, is too well known to require enumeration. To this list we must now add *Sprigs of Heather, or the Rambles of 'Mayfly' with old Friends*, by the Rev. John Anderson, D.D., Minister of Kinnoull. Mr Anderson is a veteran angler, and is able to look back to days spent by the river-side with the great Christopher North, and with others who, though of less note in the angling and literary world, were still such as to afford to the author the opportunity of telling many amusing and characteristic stories regarding them. He is, as many, perhaps most, anglers are, delighted with the scenes of rural beauty into which his pursuits have led him, and he describes them with the pen of a ready and accomplished writer, and with somewhat of poetic fervour. Mr Anderson is a strong advocate of fly-fishing, and almost scornfully speaks of those who use bait, as 'ground-fishers,' and the like. We are not sure but his indignation on this point is misplaced, as all bait-fishing is not done in muddy or discoloured water, and perhaps as much skill is required to fish successfully a small clear stream with worm as with fly. Stewart and other well-known anglers have long since acknowledged this. In other respects, however, Mr Anderson's little volume is such that lovers of the rod and line will find it entertaining reading.



Those who love Scottish music and Scottish dances will hail with pleasure the appearance of two handsome volumes entitled, *The Athole Collection of Dance Music of Scotland* (Edinburgh: Maclachlan and Stewart). These volumes have been compiled and arranged by Mr James Stewart Robertson (Edradynate), who has done his work in a most efficient manner. He, as an unprofessional musician, apologises for having undertaken such a work, which, he says, was only done by him because he did not expect, from the disfavour into which, for the present, Scottish music and dances have unfortunately fallen, that any professional musician, competent for the task, 'could be induced to devote the time, and to run the chances attending the production of such a work.' So far as Mr Robertson's execution of the work is concerned, no such apology was required; while his devotion to the task which he has so satisfactorily accomplished renders his services to his

country almost patriotic. He has selected his airs with admirable taste and skill, and the two volumes contain within them specimens of almost every characteristic of Scottish dance music. No better or more acceptable present could be sent from Scotch folks at home to Scotch folks abroad than this *Athole Collection*.

AMONG THE DAISIES.

LAY her down among the daisies,
With the fringes of her eyes,
Softer than their silver petals,
Closed for blissful reveries.
Fold her little hands in whiteness
As in prayer on her breast;
Fear not for their folded lightness
On the heart unmoving pressed,
For that heart of angel brightness,
Tired so early, lies at rest.

Tired so early!—when the dawning
Glimmered white-winged through the room,
And the skies were half awaking,
Half in fading starlit gloom,
From the heaven of the starlight
Came the angels of the dawn;
And the morning winds were sighing,
And the curtains eastward drawn,
And her sleeping face looked brighter,
And a whispering sob said—‘Gone!’

All the daisies were unfolding
In the fields, where never more
Shall the rapture of her child-life
Run in shout and laughter o’er.
Tired so early!—she has gathered
All her gladness in swift space,
She has sung her song and ended,
Childlike turning pleading face
Back to home when joys are weary—
Toward the one familiar place.

Lay her low among the daisies:
Angels knew her more than we;
They have led her home from wandering,
Tired with earthly revelry.
And above her daisied pillow
Let her simple tale be told:
Here the Lover of the lilies
Bade a little blossom fold;
He that wakes the flowers shall wake her,
White as snow, with heart of gold.

HELEN ATTERIDGE.

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FOOTNOTES:

[\[1\]](#) For an account of this interesting repository of crime, see ‘The National Album’ in *Chambers’s Journal* for October 18, 1879.

[Transcriber’s note—the following changes have been made to this text:
Page 207: Angelos to Angeles—“Los Angeles”.]

*** END OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
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